



05 — Requirements Traceability Matrix (RTM)

TASCO Insurance × VETC Growth Platform — Business & Product

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0. How to use this matrix

- Each row traces one functional requirement (doc 02) to its user stories (doc 04), code modules, HTTP endpoints, rule kinds and tests.
- Code paths** are relative to the repository root. `domain/` = `src/domain/`, `app/` = `src/application/`, `http/` = `src/adapters/http/`, `persist/` = `src/adapters/persistence/`, `integ/` = `src/adapters/integrations/`, `rules/` = `src/rules/`, `shared/` = `src/shared/`. All HTTP endpoints are declared in `src/adapters/http/routes.js` (73 API routes); `/health/*`, `/metrics`, `/api/openapi.json` and static pages are served by `src/adapters/http/app.js`.
- Test layout:**

Layer	Path	What it covers
Unit	<code>test/unit/*.test.js</code>	Rules engine (<code>rulesEngine.test.js</code>), <code>jsonLogic</code> , <code>decisionTable</code> , <code>identity</code> , <code>enrichment</code> , <code>leads</code> , <code>rating</code> , <code>contactPolicy</code> , <code>voicebot</code> , <code>crypto</code> , <code>validation</code> , <code>stores</code> , <code>schema</code>
Integration	<code>test/integration/*.test.js</code>	Services end to end on the memory store
API	<code>test/api/*.test.js</code>	HTTP contract, authentication and authorisation
Security	<code>test/security/*.test.js</code>	Security controls
PostgreSQL	<code>test/pg/*.test.js</code>	PostgreSQL adapter
Performance	<code>test/perf/load.js</code>	Load

- Test status.** At the time of writing (snapshot 7 Oct 2026) the test suite is being built in parallel. Only `test/unit/identity.test.js` and `test/unit/jsonLogic.test.js` exist. Every other test reference below is **planned** and is marked with `±`. Re-run this matrix when the suite lands.
- Build status** is taken from doc 02: B = Built, P = Partial, PI = Planned.

1. Matrix

A. Data and MDM

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-001	US-001	app/ingestionService.js	POST /api/data/ingest	—	test/integration/ingestion.test.js†, test/api/data.test.js†	B
FR-002	US-001	domain/identity.js#normalizePlate	POST /api/data/ingest, POST /api/partner/v1/quotes	—	test/unit/identity.test.js	P (E-03)
FR-003	US-001	domain/identity.js#normalizePhone	as above	—	test/unit/identity.test.js	B
FR-004	US-001	domain/enrichment.js#buildProfiles, survivors	POST /api/data/ingest	enrichment	test/unit/enrichment.test.js†	B
FR-005	US-002	domain/enrichment.js#inferExpiry, rollForward	GET /api/customers/:id	enrichment	test/unit/enrichment.test.js†	B
FR-006	US-006	domain/enrichment.js#inferCategory, rules/decisionTable.js	GET /api/customers/:id	enrichment	test/unit/enrichment.test.js†, test/unit/decisionTable.test.js†	B
FR-007	US-004	domain/enrichment.js, app/opsService.js#dqIssues, resolveDq, app/voiceService.js#finalize	GET /api/dq/issues, POST /api/dq/issues/:id/resolve	enrichment	test/integration/ops.test.js†, test/api/data.test.js†	B
FR-008	US-002	app/opsService.js#lineage	GET /api/customers/:id/lineage	—	test/api/customers.test.js†	B (E-18)
FR-009	US-003	http/routes.js (handler), domain/enrichment.js#applyDeclaredExpiry	PATCH /api/customers/:id/expiry	—	test/api/customers.test.js†	B
FR-010	US-005, US-027	app/ingestionService.js#rebuild, app/customerService.js#declareExpiry, app/voiceService.js#finalize, app/salesService.js#purchase	POST /api/customer/expiry, POST /api/voice/sessions/:id/turns	enrichment	test/integration/ingestion.test.js†	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-011	US-001	app/ingestionService.js#rebuild, bootstrap/container.js#registerSubscribers	POST /api/data/ingest	—	test/integration/ingestion.test.js†	B

B. Lead intelligence

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-012	US-007	domain/leads.js#factsFor	—	all lead kinds	test/unit/leads.test.js†	B
FR-013	US-017, US-018	domain/leads.js#assignJourney, rules/jsonLogic.js	GET /api/leads	journeys	test/unit/leads.test.js†, test/unit/jsonLogic.test.js	B
FR-014	US-007	domain/leads.js#score	GET /api/leads	scoring	test/unit/leads.test.js†	B
FR-015	US-008, US-054	domain/leads.js#evaluateLead, rules/decisionTable.js	GET /api/leads	nba	test/unit/leads.test.js†, test/unit/decisionTable.test.js†	B
FR-016	US-019	domain/leads.js#benefitsFor	GET /api/leads, GET /api/customers/:id	benefits	test/unit/leads.test.js†	B
FR-017	US-007, US-009	app/leadService.js#list, app/accessPolicy.js	GET /api/leads	abac	test/api/leads.test.js†, test/security/authz.test.js†	B
FR-018	US-011	app/leadService.js#recompute, bootstrap/container.js (subscribers), jobs/cli.js (recompute)	POST /api/leads/recompute	—	test/integration/leads.test.js†	B
FR-019	US-010	app/leadService.js#recomputeOnce, hasActiveTascoCover	—	—	test/integration/sales.test.js†	B

C. Journeys and messaging

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-020	US-012, US-017, US-019	domain/leads.js, rules/validators.js	—	journeys	test/unit/rulesEngine.test.js†	B
FR-021	US-012	domain/leads.js#planTouchpoints	GET /api/touchpoints	journeys	test/unit/leads.test.js†	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-022	US-012, US-017	app/journeyService.js#runDue, executeTouchpoint, jobs/cli.js (journeys)	POST /api/journeys/run	journeys	test/integration/journeys.test.js\$, test/api/journeys.test.js\$	B
FR-023	US-013, US-015	domain/contactPolicy.js#canContact	POST /api/journeys/run, POST /api/ecosystem/events	contact_policy	test/unit/contactPolicy.test.js\$	B (E-01)
FR-024	US-014, US-067	domain/contactPolicy.js#checkCopy, rules/validators.js#copyViolations	POST /api/rules, POST /api/journeys/run	copy_guard	test/unit/contactPolicy.test.js\$, test/unit/rulesEngine.test.js\$	B
FR-025	US-012, US-015, US-019	app/journeyService.js#sendMessage, rules/jsonLogic.js#template, bootstrap/container.js (links)	POST /api/journeys/run	content.messages	test/integration/journeys.test.js\$	B
FR-026	US-018, US-020, US-021, US-022	app/journeyService.js#handleEcosystemEvent	POST /api/ecosystem/events	triggers, contact_policy	test/integration/journeys.test.js\$, test/api/journeys.test.js\$	B (live feed PI)
FR-027	US-040	bootstrap/container.js (subscriber confirmation-and-cross-sell)	POST /api/customer/orders (side effect)	content.messages	test/integration/sales.test.js\$	B
FR-028	US-046	bootstrap/container.js (subscriber confirmation-and-cross-sell)	—	journeys.crossSell	test/integration/sales.test.js\$	B
FR-029	US-012	app/journeyService.js#schedule	GET /api/touchpoints	—	test/api/journeys.test.js\$	B
FR-030	US-010	app/journeyService.js#executeTouchpoint	POST /api/journeys/run	—	test/integration/journeys.test.js\$	B

D. AI voice bot

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-031	US-023	domain/voicebot.js#createDialogue.start	POST /api/voice/sessions	content.voicebot	test/unit/voicebot.test.js\$	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-032	US-024	domain/voicebot.js#turn (state verify_plate), domain/identity.js#extractPlateFromSpeech	POST /api/voice/sessions/:id/turns	content.voicebot	test/unit/voicebot.test.js ‡, test/unit/identity.test.js	B
FR-033	US-024	domain/voicebot.js (state confirm_expiry, maskPlate)	POST /api/voice/sessions/:id/turns	content.voicebot	test/unit/voicebot.test.js ‡	B
FR-034	US-026	domain/voicebot.js#keywordClassifier	POST /api/voice/sessions/:id/turns	content.voicebot	test/unit/voicebot.test.js ‡	B
FR-035	US-025	domain/voicebot.js (intent price)	POST /api/voice/sessions/:id/turns	content.voicebot, copy_guard	test/unit/voicebot.test.js ‡	B
FR-036	US-027, US-028	app/voiceService.js#finalize	POST /api/voice/sessions/:id/turns, POST /api/voice/campaign	—	test/integration/voice.test.js‡	B
FR-037	US-030	app/voiceService.js#start, turn, get	POST /api/voice/sessions, GET /api/voice/sessions/:id, POST /api/voice/sessions/:id/turns	—	test/api/voice.test.js‡	B
FR-038	US-029, US-054	http/routes.js (campaign handler), app/voiceService.js#autoCall, integ/simulatedCaller.js	POST /api/voice/campaign	contact_policy (should apply)	test/integration/voice.test.js‡	P (E-01, E-02)
FR-039	US-031	app/insightsService.js#governance	GET /api/dashboard/governance	—	test/api/dashboards.test.js ‡	B

E. Telesales

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-040	US-032	domain/voicebot.js#handoffSummary	GET /api/handoffs/:id	—	test/unit/voicebot.test.js‡	B
FR-041	US-033	app/voiceService.js#listHandoffs, app/accessPolicy.js#check	GET /api/handoffs, GET /api/handoffs/:id	abac	test/api/handoffs.test.js‡, test/security/authz.test.js‡	B
FR-042	US-033	app/voiceService.js#updateHandoff	PATCH /api/handoffs/:id	—	test/integration/voice.test.js‡	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-043	US-034	http/routes.js (handler), app/accessPolicy.js#require	PATCH /api/handoffs/:id	rbac	test/api/handoffs.test.js†	B
FR-044	US-016	app/voiceService.js#createDirectHandoff, app/journeyService.js	POST /api/journeys/run	journeys	test/integration/journeys.test.js†	B

F. Sales and issuance

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-045	US-035	app/salesService.js#catalogue	GET /api/products	products	test/api/sales.test.js†	B
FR-046	US-035, US-039, US-041, US-044	app/salesService.js#quote, domain/rating.js#coverStartDate	POST /api/quotes, POST /api/customer/quotes, POST /api/partner/v1/quotes	products, tariff.*, rating.*, benefits	test/integration/sales.test.js†, test/unit/rating.test.js†	B
FR-047	US-035, US-039, US-051	app/salesService.js#purchase, integ/mockGateways.js (wallet, core), shared/resilience.js	POST /api/orders, POST /api/customer/orders, POST /api/partner/v1/orders	—	test/integration/sales.test.js†, test/api/sales.test.js†	B (sandbox ports)
FR-048	US-042	app/salesService.js#purchase (compensation)	as above	—	test/integration/sales.test.js†	B
FR-049	US-039	app/salesService.js#purchase (golden-record update)	as above	—	test/integration/sales.test.js†	B
FR-050	US-040, US-077	app/salesService.js#verifyCertificate, http/app.js (/verify/* → verify.html)	GET /api/public/certificates/:certNo, /verify/:certNo	—	test/api/public.test.js†	B (page PI)
FR-051	US-035	app/salesService.js#listPolicies	GET /api/policies	—	test/api/sales.test.js†	B
FR-052	US-074	app/opsService.js#reconcile, jobs/cli.js (reconcile)	POST /api/ops/jobs/:kind (reconciliation)	—	test/integration/ops.test.js†	B

G. Products and rating

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-053	US-006, US-039, US-041	domain/rating.js#METHODS.tariff_table	POST /api/quotes (+ customer and partner variants)	tariff.tnds_car, tariff.tnds_motorbike	test/unit/rating.test.js†	B
FR-054	US-045	domain/rating.js#METHODS.rate_on_sum_insured	as above	rating.motor_p	test/unit/rating.test.js†	B (E-06)
FR-055	US-044	domain/rating.js#METHODS.per_seat	as above	rating.pa_seat	test/unit/rating.test.js†	B
FR-056	US-025, US-039	domain/rating.js#rate (priceRegulated note)	as above	products	test/unit/rating.test.js†	B
FR-057	US-063	rules/validators.js (products), domain/rating.js	POST /api/rules	products	test/unit/rulesEngine.test.js†	B

H. Value and benefits

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-058	US-043	domain/leads.js#benefitsFor, app/leadService.js#benefitsForStaff	GET /api/customer/home, GET /api/customers/:id	benefits	test/unit/leads.test.js†	B
FR-059	US-043	domain/leads.js#benefitsFor	as above	benefits	test/unit/leads.test.js†	B
FR-060	US-047	— (gated config only)	—	benefits (loyalty_points), referral	test/unit/leads.test.js† (gating)	PI
FR-061	US-048	—	— (endpoint to be added)	benefits (auto_renew)	test/integration/customer.test.js†	PI (E-10)
FR-062	US-020	app/journeyService.js#handleEcosystemEvent	POST /api/ecosystem/events	triggers, benefits	test/integration/journeys.test.js†	P

I. Partners and fleet

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-063	US-049, US-053	app/partnerService.js#create, setStatus, list	GET/POST /api/partners, PATCH /api/partners/:id	—	test/api/partners.test.js†	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-064	US-049	app/partnerService.js#issueApiKey, revokeApiKey, authenticateKey, shared/crypto.js	POST /api/partners/:id/keys, DELETE /api/partners/keys/:keyId	—	test/api/partners.test.js†, test/security/authz.test.js†	B
FR-065	US-050	http/routes.js (partner quote handler), app/ingestionService.js, app/salesService.js#quote	POST /api/partner/v1/quotes	products, enrichment	test/api/partnerApi.test.js†	B (E-11)
FR-066	US-051	http/routes.js (ownership check), app/salesService.js#purchase	POST /api/partner/v1/orders	—	test/api/partnerApi.test.js†	B
FR-067	US-052	app/salesService.js#listPolicies, app/partnerService.js#statement	GET /api/partner/v1/policies, GET /api/partner/v1/statement	—	test/api/partnerApi.test.js†	B
FR-068	US-052, US-068	domain/rating.js#commissionFor, rules/validators.js (commission)	POST /api/partner/v1/orders (side effect), POST /api/rules	commission	test/unit/rating.test.js†, test/unit/rulesEngine.test.js†	B
FR-069	US-052	app/partnerService.js#statement	GET /api/partners/:id/statement	—	test/api/partners.test.js†	B
FR-070	US-054, US-055	domain/leads.js (NBA), http/routes.js (campaign excludes company)	GET /api/leads?action=route_b2b	nba, benefits (fleet_dashboard)	test/unit/leads.test.js†	P (E-07, E-10)

J. Claims FNOL

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-071	US-056	app/claimsService.js#submit	POST /api/customer/claims	— (SLA should be a rule, E-05)	test/integration/claims.test.js†, test/api/customerApp.test.js†	B
FR-072	US-057	app/claimsService.js#transition, list (TRANSITIONS)	GET /api/claims, PATCH /api/claims/:id	—	test/integration/claims.test.js†	B
FR-073	US-058	app/claimsService.js#list	GET /api/customer/claims	—	test/api/customerApp.test.js†	B

K. Customer self-service and privacy

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-074	US-036	bootstrap/container.js (links.sign/verify), app/customerService.js#issueCustomerToken	POST /api/customer/session	abac (customer_self)	test/api/customerApp.test.js\$, test/security/authz.test.js\$	B (VETC SSO PI)
FR-075	US-037	app/customerService.js#home	GET /api/customer/home	benefits	test/integration/customer.test.js\$	B
FR-076	US-038	app/customerService.js#declareExpiry	POST /api/customer/expiry	—	test/integration/customer.test.js\$	B
FR-077	US-035, US-039	app/salesService.js	POST /api/customer/quotes, POST /api/customer/orders	products, tariff.*, rating.*	test/api/customerApp.test.js\$	B (UI PI)
FR-078	US-028, US-059	app/customerService.js#updateConsent, app/ingestionService.js#rebuild (overrides)	PUT /api/customer/consent	contact_policy	test/integration/customer.test.js\$	B
FR-079	US-060	app/customerService.js#exportData	GET /api/customer/data-export	—	test/api/customerApp.test.js\$	B
FR-080	US-061	app/customerService.js#exportData, erase	POST /api/dsar/:id/export, POST /api/dsar/:id/erase	retention	test/integration/customer.test.js\$	P (E-19)
FR-081	US-062	app/accessPolicy.js#maskProfile, shared/util.js#maskPhone, maskName	GET /api/customers/:id	rbac	test/security/authz.test.js\$	B

L. Rules governance

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-082	US-063	app/rulesService.js#list, byId, get	GET /api/rules, GET /api/rules/:id	all	test/integration/rules.test.js\$	B
FR-083	US-063, US-067, US-068	app/rulesService.js#createDraft, rules/validators.js, rules/jsonLogic.js#validate, rules/decisionTable.js#validateTable	POST /api/rules	all	test/unit/rulesEngine.test.js\$, test/unit/jsonLogic.test.js, test/unit/decisionTable.test.js\$	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-084	US-063	app/rulesService.js#validate	POST /api/rules/validate	all	test/api/rules.test.js	B
FR-085	US-064	http/routes.js (simulate handler), domain/leads.js#evaluateLead	POST /api/rules/simulate	scoring , nba , journeys , benefits	test/api/rules.test.js	B
FR-086	US-065	app/rulesService.js#submit , approve , reject	POST /api/rules/:id/submit , /approve , /reject	all	test/integration/rules.test.js , test/security/authz.test.js	B
FR-087	US-066	app/rulesService.js#rollback	POST /api/rules/:id/rollback	all	test/integration/rules.test.js	B
FR-088	—	app/rulesService.js#loadDefaults , bootstrap/seed.js#seedRules	— (CLI npm run job rules)	all	test/integration/rules.test.js	B

M. Identity and access

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-089	US-069, US-070	app/identityService.js#login , shared/crypto.js#hashPassword , verifyPassword	POST /api/auth/login	—	test/integration/identity.test.js , test/api/auth.test.js	B
FR-090	US-069	app/identityService.js#verifyMfa , shared/crypto.js#totp , verifyTotp	POST /api/auth/mfa , GET /api/demo/totp/:username (demo only)	—	test/unit/crypto.test.js , test/api/auth.test.js	B
FR-091	US-069	shared/crypto.js#signJwt , verifyJwt , app/identityService.js#logout , authenticate	POST /api/auth/logout , GET /api/auth/me	—	test/unit/crypto.test.js , test/security/authz.test.js	P (E-15)
FR-092	US-071	app/accessPolicy.js#permissions , require	all staff routes	rbac	test/security/authz.test.js	B
FR-093	US-009, US-033	app/accessPolicy.js#check	GET /api/customers/:id , GET/PATCH /api/handoffs*	abac	test/security/authz.test.js	P (E-18)
FR-094	US-071	app/identityService.js#createUser , update , list	GET/POST /api/users , PATCH /api/users/:id	rbac	test/api/users.test.js	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-095	US-036, US-049	app/partnerService.js#authenticateKey, app/identityService.js#authenticate, http/app.js	partner and customer routes	—	test/security/authz.test.js‡	B
FR-096	US-071	app/identityService.js#changePassword, shared/crypto.js#checkPasswordPolicy	POST /api/auth/password	—	test/unit/crypto.test.js‡	B

N. Audit

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-097	US-003, US-062	app/auditService.js#record (called by all services), persist/memoryStore.js, persist/postgresStore.js	—	—	test/integration/*.test.js‡	B
FR-098	US-072	app/auditService.js#list	GET /api/audit	—	test/api/audit.test.js‡	B
FR-099	US-072	persist/auditChain.js#verifyChain, entryHash, canonical, db/migrations/001_init.sql (append-only trigger)	GET /api/audit/verify	—	test/unit/stores.test.js‡, test/pg/postgresStore.test.js‡	B

O. Operations, reporting and UI

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-100	US-073	app/insightsService.js#overview	GET /api/dashboard/overview	costs	test/integration/insights.test.js‡, test/api/dashboards.test.js‡	B (E-12, E-16)
FR-101	—	app/insightsService.js#adoption	GET /api/dashboard/adoption	—	test/api/dashboards.test.js‡	B
FR-102	US-031	app/insightsService.js#governance	GET /api/dashboard/governance	—	test/api/dashboards.test.js‡	B
FR-103	US-074	http/routes.js (ops status), shared/resilience.js	GET /api/ops/status	—	test/api/ops.test.js‡	B
FR-104	US-074	app/opsService.js#recordRun, jobRuns, adapters/messaging/outboxEventBus.js, jobs/cli.js	GET /api/ops/jobs, POST /api/ops/jobs/:kind	—	test/integration/ops.test.js‡	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-105	US-075	app/opsService.js#applyRetention	POST /api/ops/jobs/retention	retention	test/integration/ops.test.js‡	P (E-09)
FR-106	—	http/app.js , http/openapi.js , shared/metrics.js	/health/live , /health/ready , /metrics , /api/openapi.json , GET /api/meta	—	test/api/platform.test.js‡	B
FR-107	—	bootstrap/seed.js , integ/syntheticVetcsSource.js , shared/config.js	GET /api/demo/totp/:username , GET /api/meta (demo customers)	—	test/integration/seed.test.js‡	B
FR-108	US-022, US-076	public/index.html , public/js/console/* , public/css/*	all staff routes	—	UI end-to-end (planned, tool TBD)	PI
FR-109	US-077	public/app/index.html , public/js/customer/* , public/verify.html , public/js/verify.js	customer and public routes	—	UI end-to-end (planned)	PI
FR-110	US-076	public/js/shared/* (i18n, theme)	—	—	UI end-to-end (planned)	PI

2. Coverage summary

Capability	FRs	With ≥ 1 user story	With API endpoint	With rule kind	Built	Partial	Planned
A Data and MDM	11	11	11	5	10	1	0
B Lead intelligence	8	8	6	6	8	0	0
C Journeys and messaging	11	11	9	9	11	0	0
D Voice bot	9	9	9	6	8	1	0
E Telesales	5	5	5	3	5	0	0
F Sales and issuance	8	8	8	2	8	0	0
G Products and rating	5	5	5	5	5	0	0
H Value and benefits	5	5	3	5	2	1	2
I Partners and fleet	8	8	8	3	7	1	0
J Claims	3	3	3	0	3	0	0
K Customer and privacy	8	8	8	6	7	1	0
L Rules governance	7	6	6	7	7	0	0
M Identity and access	8	8	8	3	6	2	0
N Audit	3	3	2	0	3	0	0

Capability	FRs	With ≥ 1 user story	With API endpoint	With rule kind	Built	Partial	Planned
O Ops and UI	11	8	10	2	7	1	3
Total	110	106	101	62	97	8	5

Counting rule: "Built" here counts B and B(...) rows (built with a noted gap or a planned UI or integration), "Partial" counts P rows, and "Planned" counts PI rows. FR-088, FR-101, FR-106 and FR-107 are platform requirements without a dedicated user story.

3. Engineering notes: inconsistencies found during documentation

These were found while tracing requirements to code. They are recorded here so that engineering can triage them; IDs are referenced from docs 01–04. **Severity:** H = compliance, security or money risk; M = functional gap or misleading behaviour; L = hygiene.

ID	Sev	Finding	Evidence	Recommendation
E-001	H	The automated voice campaign route bypasses the contact policy. It checks only <code>consent.call</code> , <code>dnc</code> , phone and owner type, so it ignores the 08:00–20:00 window, the weekly call cap and marketing consent.	<code>src/adapters/http/routes.js</code> (<code>POST /api/voice/campaign</code>) vs <code>domain/contactPolicy.js#canContact</code>	Call <code>canContact(policy, profile, 'voice_bot', { marketing: true, now, history })</code> per lead and report skipped reasons.
E-002	M	<code>voiceService.autoCall</code> does not itself refuse DNC customers (<code>start()</code> does); it relies on callers.	<code>src/application/voiceService.js</code>	Guard DNC and consent inside <code>autoCall</code> .
E-003	M	Plate regex accepts only car formats (<code>^\d{2}[A-Z]{1,2}\d{4,5}\$</code>). The comment says motorbike plates like <code>51G1-678.90</code> are handled, but they are rejected, so <code>TNDS_MOTORBIKE</code> cannot be sold to most motorbikes. Lead premium is always computed with the <code>tariff.tnds_car</code> rule.	<code>src/domain/identity.js</code> , <code>src/application/leadService.js#tndsPremium</code>	Extend the regex to series + digit (e.g. <code>G1</code>); choose the tariff by vehicle class.
E-004	M	Product channel <code>zalo</code> exists in <code>products.json</code> , but no route ever sets <code>channel: 'zalo'</code> . Customer quotes are hardcoded to <code>vetc_app</code> , so Zalo mini-app sales cannot be attributed or restricted.	<code>routes.js</code> (<code>POST /api/customer/quotes</code>)	Derive the channel from the customer token's audience or client ID.

ID	Sev	Finding	Evidence	Recommendation
E-05	M	Business thresholds are hardcoded, contrary to NFR-062: • claim SLA 4 h (<code>claimsService.js</code>) • quote TTL 24 h (<code>salesService.js</code>) • <code>needsConfirmation < 0.75</code> (<code>customerService.js</code>) • <code>expiryKnown ≥ 0.5</code> (<code>voicebot.js</code> , should read <code>enrichment.usableExpiryConfidence</code>) • customer-declared confidence 0.8 (<code>customerService.js</code>) vs 0.75 in <code>enrichment.json</code> • steward confidence 0.9 (<code>routes.js</code>) • bot "already renewed" confidence 0.6 and +365 days (<code>voiceService.js</code>) • stuck-order threshold 1 h (<code>opsService.js</code>) • DQ completeness denominator 5 (<code>enrichment.js</code>)	as listed	Move these into rule kinds (<code>enrichment</code> , a new <code>sales / claims</code> kind) and validate them.
E-06	M	MOTOR_PD has <code>requiresInspection: true</code> , but quote and purchase never enforce a vehicle inspection or photo step, and it is sellable on <code>vetc_app</code> .	<code>products.json</code> , <code>salesService.js</code>	Add a pre-bind inspection state (photo upload or assessor) or remove PD from self-service channels until ready.
E-07	M	Company-owned vehicles enter B2C journeys: journey audiences do not exclude <code>ownerType = company</code> , while the NBA routes them to B2B. Fleet drivers can therefore get B2C reminders and calls from journey steps.	<code>journeys.json</code> , <code>nba.json</code>	Add <code>{"!=": [{"var": "ownerType"}, "company"]}</code> to the B2C audiences, or add a <code>fleet</code> journey.
E-08	L	The <code>hot_call</code> NBA recommends <code>voice_bot</code> on <code>consent.call</code> alone. Execution also requires <code>marketing consent</code> (marketing step), so the NBA can recommend an action the contact policy will block.	<code>nba.json</code> , <code>contact_policy.js</code> on	Add <code>consent.marketing</code> to <code>hot_call</code> .
E-09	M	Retention is only partly executed: only <code>source_records</code> and <code>voice_sessions</code> are deleted. Profile anonymisation after 1,825 days, and archival of messages, orders, certificates and audit, are reported as "executed by archival pipeline", which does not exist.	<code>opsService.js#app</code> <code>lyRetention</code>	Implement anonymisation and archival, or mark them clearly as manual in the ops UI.
E-10	M	Customer-visible approved benefits promise features that do not exist yet: <code>auto_renew</code> (no opt-in endpoint or confirm-before-debit flow) and <code>fleet_dashboard</code> (no fleet view or consolidated invoice). The NBA label and the <code>/api/customer/expiry</code> summary say "earns points" while <code>loyalty_points</code> is pending legal review.	<code>benefits.json</code> , <code>nba.json</code> (<code>fix_data</code>), <code>routes.js</code>	Set these benefits to <code>pending</code> until built; remove the "earns points" wording until loyalty is approved.

ID	Sev	Finding	Evidence	Recommendation
E-11	M	Partner API: <code>ownerType</code> is hardcoded to <code>individual</code> , so fleet partners cannot onboard company vehicles. Any active partner can quote any plate in the VETC base, and partner-supplied <code>currentExpiry</code> is ingested as evidence, which is a data-poisoning and enumeration vector.	<code>routes.js</code> (<code>POST /api/partner/v1/quotes</code>)	Accept <code>ownerType</code> ; rate-limit per key; weight partner evidence by source trust (already the case) and flag conflicts as DQ issues; consider a consent token for existing VETC customers.
E-12	L	The dashboard metric <code>profilesWithUsableData</code> uses <code>dq_score >= 50</code> as a proxy, not <code>expiryConfidence >= usableExpiryConfidence</code> (the K-01 definition).	<code>insightsService.js#overview</code>	Index <code>policy.expiryConfidence</code> and count against the rule threshold.
E-13	L	<code>copy_guard.appliesTo: "priceRegulated"</code> is ignored: the guard runs on all customer copy. This is conservative, but the configuration is misleading.	<code>copy_guard.json</code> , <code>contactPolicy.js#checkCopy</code>	Document "all copy" or implement product scoping.
E-14	L	Trigger <code>wallet_topup</code> fires for days <code>-30..30</code> but uses <code>first_reminder</code> ("expires on <code>{{expiry}}</code> "). For lapsed vehicles that reads as a future date.	<code>triggers.json</code>	Use <code>lapsed_notice</code> when <code>days < 0</code> (split into two triggers).
E-15	M	Logout revocation and rate limiting are held in memory per replica, so they are ineffective with more than one replica.	<code>identityService.js</code> (<code>revoked Map</code>), <code>security.js#createRateLimiter</code>	Move to a shared store (Postgres table or Redis) or enforce at the gateway or WAF.
E-16	L	Insights do not scale to 6 M records: <code>adoption()</code> reads only the latest 5,000 audit entries, and <code>overview()</code> pages through all orders to sum premium.	<code>insightsService.js</code>	Use SQL aggregates or materialised views.
E-17	L	Phone masking is inconsistent: <code>createDirectHandoff</code> builds its own mask instead of <code>shared/util.js#maskPhone</code> .	<code>voiceService.js</code>	Use the shared helper.
E-18	M	ABAC regional policy is not applied to <code>GET /api/customers/:id/lineage</code> or <code>PATCH /api/customers/:id/expiry</code> . <code>GET /api/handoffs/:id</code> lets an agent read unassigned handoffs from other regions by ID.	<code>routes.js</code>	Call <code>access.check(principal, 'read', {type: 'profile', region})</code> on lineage and expiry; add a region condition to <code>agent_own_handoffs</code> .
E-19	H	DSAR erasure does not scrub PII held in <code>handoffs</code> (name), <code>claims</code> (description, location) or audit <code>details</code> .	<code>customerService.js#erase</code> , <code>schema.js</code>	Extend erasure to these collections; keep audit entries but pseudonymise details.
E-20	M	Quality gates are not yet runnable: <code>test/</code> is being built (2 unit files at snapshot), so coverage thresholds in <code>package.json</code> cannot pass. There is no ESLint config, Dockerfile or K8s manifest yet. Docs referenced in code (<code>docs/architecture/integration-architecture.pdf</code> , ADR-004/005/007) must exist.	repository root	Track in the delivery plan; block release on NFR-042 and NFR-043.

ID	Sev	Finding	Evidence	Recommendation
E-21	L	Rule simulation supports only <code>scoring</code> , <code>nba</code> , <code>journeys</code> and <code>benefits</code> , not tariffs, contact policy or triggers.	<code>routes.js</code> (simulate)	Extend to quote simulation and <code>canContact</code> dry-runs.
E-22	H	Staff-initiated purchase (<code>POST /api/orders</code> with <code>policy:issue</code>) debits the customer's VETC wallet without in-app customer confirmation. This contradicts the trust principle "payment only in the VETC app" and the bot and telesales scripts.	<code>routes.js</code> , <code>salesService.js#purchase</code>	Require a customer confirmation (in-app approve or wallet pre-authorisation) for staff-channel orders, or restrict <code>POST /api/orders</code> to sending a pay link.